

Parkinson Disease Prediction using Machine Learning In Healthcare

Ansh Maheshwari
Department of Computing
Technologies
SRM Institute of Science and
Technology
Kattankulathur, India
am2328@srmist.edu.in

Ishita Banerjee
Department of Computing
Technologies
SRM Institute of Science and
Technology
Kattankulathur, India
ib0342@srmist.edu.in

Dr. Ramkumar Jayaraman
Department of Computing
Technologies
SRM Institute of Science and
Technology
Kattankulathur, India
ramkumaj@srmist.edu.in

Abstract— Detecting Parkinson's Disease at early stages remains difficult because the symptoms begin slowly at first. The investigation analyzes how machine learning (ML) strengthens diagnostic precision by using vocal biomarkers. We have applied supervised learning algorithm assessment on public voice data by using Random Forest along with ensemble models while performing standard classification measurements. Random Forest using Entropy as well as Voting Classifier provided the most effective results for identifying participants with PD among those without PD. The study demonstrates that ML-based voice analysis represents an accessible and non-obtrusive method to help clinicians make early PD diagnoses and make clinical decisions.

Keywords— *Parkinson's Disease, Machine Learning, Voice-Based Biomarkers, Classification Algorithms, Random Forest, Voting Classifier, Supervised Learning*

I. INTRODUCTION

Parkinson's Disease functions as a persistent condition that develops slowly toward degeneration while targeting the motor operations including movement and balance alongside coordination. This condition ranks as the second highest neurological condition worldwide after Alzheimer's disease. The key Parkinson's Disease symptoms consist of tremors as well as muscle stiffness in combination with bradykinesia (slowness of movement) and postural instability. Successful patient life quality depends heavily on making Parkinson's Disease diagnosis early so treatment becomes more effective. The existing diagnostic approaches mainly use subjective evaluations in clinical exams which lead to either delayed or incorrect disease identification.

Automated diagnostic equipment can be developed with the modern breakthroughs in Artificial intelligence and machine learning technologies. These diagnostic approaches depend on biomarkers that include speech characteristics together with respiration indicators alongside retinal picture analysis to offer quick and precise and minimally invasive diagnostic testing. Different research projects investigating PD detection through ML demonstrated valuable findings yet they each present particular hindrance. The sparse availability of multiclass classification data and the requirement of unique hardware for nocturnal breathing signal analysis and complicated deep learning models for retinal image interpretation reduces both model capability to generalize and ease of interpretation.

Vocal biomarkers represent one of multiple approaches that serve as practical screening tools which are accessible and affordable for diagnosis purposes. Measurements of jitter together with shimmer and harmonics-to-noise ratio

demonstrate high potential for detecting panic disorder while their effectiveness depends on the environmental noise levels along with the recording system capabilities.

The main research objective of this study is to create an approachable and flexible yet reliable machine learning system which detects PD through biomedical voice assessment. This work evaluates multiple supervised learning algorithms consisting of Random Forest and K-Nearest Neighbors together with Voting Classifiers ensemble methods. An analysis of publicly accessible vocal recordings of people with and without Parkinson's disease will determine the most efficient classification model.

The objectives of this paper are as follows:

- 1) Researchers need to study the effectiveness of ML algorithms in identifying Parkinson's disease through speech-based characteristics.
- 2) To compare the performance of individual and ensemble classifiers.
- 3) To assess the practicality of using voice analysis as a non-invasive diagnostic tool.

The research describes the analytical methods followed by a presentation of algorithmic performance data and addresses the future applications for producing functional ML-based tools to identify PD at an early stage:

- Related works
- Problem definition
- System Model
 - Random Forest Classifier
 - K-Nearest Neighbours Classifier
- Results
- Conclusion

II. RELATED WORKS

In the last decade, a number of research works have investigated machine learning and artificial intelligence methods for Parkinson's Disease detection from various biomarkers. These methods use data in the forms of voice recordings, handwriting, gait pattern, and medical images to develop predictive models. The following is an exhaustive summary of some of the most pertinent works in this direction.

1. Machine Learning-Based Parkinson's Disease Prediction

[1] Zhang et al., 2023 – Biomolecules

This study applied various machine learning (ML) techniques to predict Parkinson's Disease (PD) from biomedical data. Comparisons were made between models like SVM, Random Forest, and logistic regression using clinical features. This study highlighted the fact that preprocessing of data and feature selection play a significant role in affecting the precision of models. Their results illustrated the potential of ML in supporting early diagnosis of PD.

Reference: J. Zhang, W. Zhang, H. Zhang, et al., "Prediction of Parkinson's disease using machine learning methods," *Biomolecules*, vol. 13, no. 12, p. 1761, 2023.

2. Structural MRI Characteristics in Parkinson's Diagnosis

[2] Ji et al., 2022 – Frontiers in Aging Neuroscience

Ji et al. used machine learning on various structural MRI characteristics for PD diagnosis. The work illustrated the power of imaging biomarkers in differentiating PD and non-PD patients. Texture and volumetric features were utilized to train models such as SVM and decision trees. Their model yielded high classification accuracy and clinical interpretability. This is indicative of the feasibility of MRI-based ML diagnosis.

Reference: Y. Ji, J. Zou, J. Zhang, et al., "Machine learning models for diagnosis of Parkinson's disease using multiple structural magnetic resonance imaging features," *Front. Aging Neurosci.*, vol. 14, p. 633752, 2022.

3. ML Literature Review on Parkinson's Diagnosis

[3] Mei et al., 2021 – Frontiers in Aging Neuroscience

This review compared current machine learning research for the diagnosis of Parkinson's Disease. The authors grouped methods according to input data like voice, imaging, and sensor signals. It emphasized typical pitfalls such as data imbalance and generalizability between datasets. The paper provides a thorough overview of algorithm performance. It is a starting point for future PD research.

Reference: J. Mei, C. Desrosiers, and J. Frasnelli, "Machine learning for the diagnosis of Parkinson's disease: A review of literature," *Front. Aging Neurosci.*, vol. 13, p. 633752, 2021.

4. Systematic Review of ML Methodologies in PD

[4] Tabashum et al., 2024 – JMIR Medical Informatics

Tabashum et al. had written a systematic review of machine learning techniques in the diagnosis of Parkinson's Disease. The paper compared model accuracy, sensitivity, and feature engineering techniques across various papers. It also pointed out model validation, diversity in data type, and reproducibility. The authors pointed out increasing interest in

multimodal methods. The review is critical in directing future algorithm development.

Reference: T. Tabashum, et al., "Machine learning models for Parkinson disease: Systematic review," *JMIR Med. Inform.*, vol. 12, p. e50117, 2024.

5. Systematic review of ML in PD on arXiv

[5] Mei et al., 2020 – arXiv preprint

In this initial systematic review, authors summarized efforts at ML-based PD diagnosis on various data modalities. The paper recorded more than 100 publications and mapped feature types, classifier types, and evaluation metrics. It also examined publication trends and dataset limitations. In this research, gaps and opportunities in existing research were revealed.

Reference: J. Mei, C. Desrosiers, and J. Frasnelli, "Machine learning for the diagnosis of Parkinson's disease: A systematic review," *arXiv preprint, arXiv:2010.06101*, 2020.

6. Vocal Feature Deep Learning for PD Detection

[6] Gunduz, 2019 – IEEE Access

This work utilized deep learning algorithms that utilized vocal features to classify Parkinson's Disease patients. Convolutional Neural Networks (CNNs) were applied to dysphonic features extracted from audio recordings. The performance was shown to be much better than conventional classifiers. Gunduz proved the power of deep learning to utilize non-invasive data for precise classification of Parkinson's Disease.

Reference: H. Gunduz, "Deep learning-based Parkinson's disease classification using vocal feature sets," *IEEE Access*, vol. 7, pp. 115540–115551, 2019.

7. SVM and L1-Norm Feature Selection for Voice Data

[7] Haq et al., 2019 – IEEE Access

Haq and others have suggested a new Parkinson's detection model based on L1-norm SVM-based feature selection. Speech recordings and dysphonia measurements were considered for improving the accuracy of the model. Their approach outperformed the other voice-based approaches in accuracy and sensitivity. It highlighted the significance of appropriate feature selection in medical diagnosis.

Reference: A. U. Haq, J. P. Li, M. H. Memon, et al., "Feature selection based on L1-norm support vector machine and effective recognition system for Parkinson's disease using voice recordings," *IEEE Access*, vol. 7, pp. 37718–37734, 2019.

8. Sparse Auto-Encoders for Multimodal Learning

[8] Li et al., 2019 – ISBI Conference

This article introduced a stacked sparse auto-encoder for PD diagnosis with multimodal and longitudinal data. The approach merged various biomedical signals to facilitate

model generalizability. The technique performed better than the conventional single-modality classifiers. This article illustrates how deep structures can deal with high-dimensional complicated medical data.

Reference: S. Li, H. Lei, F. Zhou, et al., "Longitudinal and multi-modal data learning for Parkinson's disease diagnosis via stacked sparse auto-encoder," in *IEEE ISBI*, pp. 384–387, 2019.

9. Penalized Learning on Multimodal MRI

[9] Liu et al., 2016 – *Journal of Neuroscience Methods*

Liu et al. employed folded concave penalized learning to identify Parkinson's disease biomarkers from multimodal MRI data sets, improving classification performance without sacrificing interpretability. Sparsity in features was enforced to prevent overfitting risk inherent in high-dimensional MRI data. The work extends the application of interpretable machine learning to neuroimaging.

Reference: H. Liu, G. Du, L. Zhang, et al., "Folded concave penalized learning in identifying multimodal MRI marker for Parkinson's disease," *J. Neurosci. Methods*, vol. 268, pp. 1–6, 2016.

10. GA-ELM Approach to Structural MRI Data

[10] Pahuja & Nagabhushan, 2016 – *CCIP Conference*

In this paper, a hybrid method involving Extreme Learning Machine (ELM) and Genetic Algorithm (GA) was presented for PD classification of T1-weighted MRI data. The model outperformed conventional methods. It showed the value of optimization-based ML in medical image processing applications. Their research bridges the gap between evolutionary computing and neural networks.

Reference: G. Pahuja and T. N. Nagabhushan, "A novel GA-ELM approach for Parkinson's disease detection using brain structural T1-weighted MRI data," in 2016 *CCIP*, pp. 1–6.

11. Sensor Data from Wearable for Tremor Detection

[11] Papadopoulos et al., 2019 – *IEEE J-BHI*

This work utilized deep multiple-instance learning with wearables' IMU data in unstructured settings. Detection of Parkinsonian tremor from non-clinical settings was the objective. The method showed robustness against ambient noise and field variations. It is an insight into the increasing criticality of wearables in monitoring PD.

Reference: A. Papadopoulos, K. Kyritsis, L. Klingelhofer, et al., "Detecting Parkinsonian tremor from IMU data collected in-the-wild using deep multiple-instance learning," *IEEE J. Biomed. Health Inform.*, vol. 24, no. 9, pp. 2559–2569, 2019.

12. Discriminating Parkinsonism with SWI and SVM

[12] Haller et al., 2013 – *European Radiology*

Haller et al. applied susceptibility-weighted imaging (SWI) using SVM to differentiate Parkinson's Disease from parkinsonian disorders. The outcome gave encouraging accuracy in early detection. It offers a non-invasive solution over more costly processes.

Reference: S. Haller, S. Badoud, D. Nguyen, et al., "Differentiation between Parkinson disease and other forms of Parkinsonism using support vector machine analysis of susceptibility-weighted imaging (SWI): Initial results," *Eur. Radiol.*, vol. 23, no. 1, pp. 12–19, 2013.

13. Voice Data Classification Comparison

[13] Islam et al., 2014 – *ICIEV*

Conference Islam et al. performed a comparison of the performance of some classifiers for PD detection using dysphonia-based voice features. SVM, ANN, and decision trees were the models they considered. Their findings pointed out the reality that the performance of the classifier significantly relies on the dataset and the features used.

Reference: M. S. Islam, I. Parvez, D. Hai, and P. Goswami, "Performance comparison of heterogeneous classifiers for detection of Parkinson's disease using voice disorder (dysphonia)," in 2014 *ICIEV*, pp. 1–7.

14. SPECT Imaging Automated PD Diagnosis

[14] Illán et al., 2012 – *Medical Physics*

This investigation suggested an automated PD diagnosis system employing DaTSCAN SPECT imaging. Feature analysis and extraction were done using image processing and classification techniques. The technique showed good diagnostic accuracy and clinical utility. It shows the feasibility of imaging-based automated diagnosis systems.

Reference: I. A. Illán, J. M. Górriz, J. Ramírez, et al., "Automatic assistance to Parkinson's disease diagnosis in DaTSCAN SPECT imaging," *Med. Phys.*, vol. 39, no. 10, pp. 5971–5980, 2012.

15. Clustering-Based Decision Support Using SVM

[15] Peker, 2016 – *Journal of Medical Systems*

Peker suggested a hybrid decision support system for clinical diagnosis using k-medoids clustering and SVM. The system used weighted attributes to give better classification. It was also applied successfully to Parkinson's disease and other diseases. The task emphasizes explainability and efficiency in ML-based clinical tools.

Reference: M. Peker, "A Decision Support System to Improve Medical Diagnosis Using a Combination of k-medoids Clustering Based Attribute Weighting and SVM," *J. Med. Syst.*, vol. 40, no. 4, p. 116, 2016.

The aforementioned work points to a growing interest in artificial intelligence-based, non-invasive diagnostic technologies for Parkinson's Disease. Notwithstanding obstacles, such as limited datasets, patient heterogeneity of clinical presentation, hardware dependence, and model interpretability, our work builds upon this initial research by means of a comparative assessment of traditional machine learning models on a vocal dataset with feature-rich inputs and

ensemble methods to improve predictive accuracy and interpretability.

III. PROBLEM DEFINITION

Parkinson's Disease (PD) is a neurodegenerative condition with a complex range of motor and non-motor signs and symptoms, of which tremor, rigidity, bradykinesia, and postural instability are of significant significance. Early diagnosis of PD in the course of the disease is an important clinical issue due to the subtlety and heterogeneity of presenting signs and symptoms and the tendency towards delayed or incorrect diagnoses with conventional clinical testing. Present methods of diagnosis are predominantly reliant on expert subjective opinion and expensive imaging technology and are therefore less accessible in resource-poor settings.

Current advances in artificial intelligence (AI) and machine learning (ML) provide encouraging avenues for designing objective, low-cost, and scalable early detectors for Parkinson's Disease. But most of the current ML-based solutions are tied to dataset diversity, hardware reliance, environmental sensitivity, or lack of model interpretability, restraining their practical usage. Methods relying on multimodal data like gait, handwriting, retinal scans, and respiratory signals have been encouraging but are mostly hindered by specialized hardware necessity or data heterogeneity and computational costs.

This research answers the call for a low-cost, non-invasive, and accurate diagnostic device based on voice-based biomarkers for detecting Parkinson's disease. The methodological framework critically assesses the performance of different conventional machine learning algorithms and ensemble methods using a publicly distributed voice dataset. The focus is on identifying models with high diagnostic performance, stability, and wide applicability and the capability of ensemble learning to enhance prediction performance. By focusing on the utilization of easily accessible voice data and conventional machine learning best practices, this research aims to encourage the development of practical, AI-based diagnostic devices that enable early intervention and enhanced patient care.

IV. SYSTEM MODEL

This research adopts a machine learning-oriented approach to classify Parkinson's Disease (PD) from biomedical vocal features. The main classification framework includes two different model—Random Forest, a type of ensemble learning algorithm, and K-Nearest Neighbours (KNN), an instance-based learning algorithm. These classifiers are selected due to their complementary strengths: scalability and robustness of Random Forest, and flexibility and simplicity of KNN to fit local patterns of data. The system is trained and tested on a labeled vocal biomarker dataset for predicting PD status. This section outlines the architectural and algorithmic frameworks suitable for each classifier.



Fig1. System Architecture Diagram

A. Random Forest Classifier

The Random Forest is a bagging ensemble learning algorithm relying on the bootstrap aggregation concept that takes the mean of a group of decision trees' predictions to produce a more stable and less noisy output. Every tree within the ensemble is trained on a unique subset of the dataset with replacement sampling, and applies a randomized feature selection algorithm to node-splitting. This introduces diversity among the base learners and avoids overfitting, which is a common flaw in decision tree classifiers.

Random Forest is a strong ensemble classification method that builds numerous decision trees while training and outputs an output class label that is a function of the mode of predictions of the individual trees. This kind of ensemble approach minimizes the risk of overfitting and increases the model's ability for generalization and thus renders the model highly suitable to medical diagnosis-type problems.

In this study, the Random Forest algorithm employs Entropy as the splitting metric, a value that assesses the impurity or disorder present in a dataset and therefore directs the tree in choosing the most salient features to split the data space. The algorithm takes advantage of the high-dimensional nature of the voice dataset to uncover intricate, non-linear patterns between features that a more straightforward classifier would not be able to identify.

The classifier uses two simple mechanisms to promote diversity between the trees:

- Bootstrap Aggregation (Bagging): Random subsets of training data are sampled with replacement.
- Random Feature Selection: At every node, only a subset of features is taken into account for splitting.

In this study, the Random Forest model is developed based on Entropy as the node split criterion to effectively manage the uncertainty in biomedical signals.

Given a dataset $D = \{(x_1, y_1), \dots, (x_n, y_n)\}$, where each $x_q \in \mathbb{R}^m$ is a feature vector and $y_i \in \{0,1\}$ is the class label, the Random Forest algorithm proceeds as follows:

- 1) Initialize number of trees T
- 2) For each tree $t = 1$ to T
 - From D , create a bootstrap sample D_t
 - Construct a decision tree h_t using:
 - For every split, a random subset of features.
 - Entropy-based information Gain for node splitting
- 3) Prediction:
 - For an instance x , each tree h_t provides a prediction $h_t(x)$
 - The final prediction is the majority vote among all trees:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \dots, h_T(x)\}$$

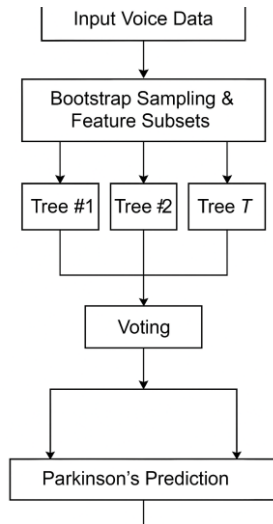


Fig 2. System Architecture Diagram – Random Forest Classifier

Random Forest possesses certain strengths if employed for the identification of Parkinson's Disease:

- 1) Generalization Capability: Reduces overfitting due to ensemble averaging.
- 2) Interpretability: Voice biomarker priority rankings help determine important voice biomarkers.
- 3) Noise Tolerance: Resistant to outliers and noisy features.
- 4) Scalability: Handles large, high-dimensional datasets well.

B. K-Nearest Neighbours (KNN) Classifier

K-Nearest Neighbours (KNN) is an instance-based, non-parametric classifier based on the proximity principle in feature space. In contrast to algorithms that build an explicit internal representation, KNN postpones all computation until query time and is thus a lazy learner. The algorithm predicts a class label for an unknown instance by the majority label of its K .

K nearest training examples, as determined by some distance measure like Euclidean distance.

KNN performs very well in low-dimensional feature spaces and for data sets with non-linear class boundaries. For PD detection from voice data, KNN provides flexible and local decision boundaries that can detect subtle variations in vocal patterns.

K nearest neighbours in feature space. It does not have an explicit training phase, but is founded upon distance computation at inference. The simplicity of KNN lends it strong baseline model potential, especially well-suited for capturing the local data structures.

For a point $x_q \in \mathbb{R}^m$ and a training dataset D , the classification process is as follows

1. Calculate the Distance:

$$d(x_q, x_i) = \sqrt{\sum_{j=1}^m (x_{qj} - x_{ij})^2}, \quad \forall i \in \{1, \dots, n\}$$

2. Find the nearest neighbours:

Identify the K samples with the smallest distances to x_q .

3. Based on the majority label among the K nearest neighbours, assign the class:

$$\hat{y} = \arg \max_{y \in \{0,1\}} \sum_{i \in \text{Neighbors}} \mathbb{I}(y_i = y)$$



Fig 3. System Architecture Diagram – KNN Classifier

KNN provides unique strengths for early-detection of PD:

- 1) Model-Free Nature: No form of distribution of the data is assumed.
- 2) Responsiveness to Localized Patterns: Recognizes subtle variations in voice that can signal the development of neurological disorders.
- 3) Simplicity: Easy to implement and interpret.
- 4) Flexibility: Simple to accommodate differences in feature scaling and class imbalance.

However, KNN is computationally costly to predict and is prone to decreased performance when handling high-dimensional data, requiring careful preprocessing and dimensionality reduction.

V. RESULTS

After rigorous testing with various models, as shown in the following figures, we have come to the conclusion that both Random Forest and K-Nearest Neighbours classifiers are the most suitable for our Parkinson's Disease Detection model due to the two models providing the most accurate results out of every one we have tested and for the features they provide.

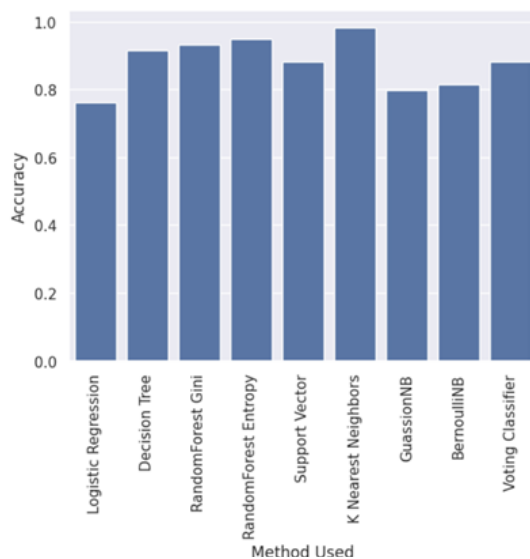


Fig 4. Different Model Accuracies Bar Chart

Figure 4 illustrates the accuracy of each classifier that we have tested for this, from Logistic Regression to Voting Classifier. This was done by comparing each accuracy percentage in a bar graph. Among them all, the KNN classifier showed the most accuracy, followed closely by Random Forest and Voting Classifier.

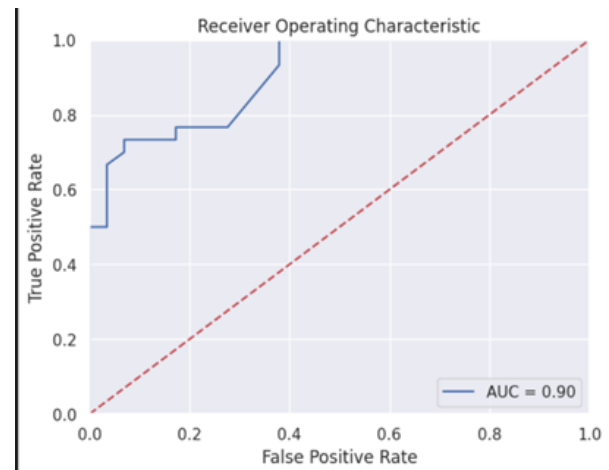


Fig 5. Receiver Operating Characteristic Graph

Figure 5 is a graph that illustrates the Receiver Operating Characteristic Curve. ROC curve is used to evaluate the overall classification performance of the model. A 0.90 AUC indicates that the model is performing well in discriminating between the diagnosed and the not diagnosed with Parkinson's disease. A high AUC indicates that the model has high discriminatory power, which is important for successful diagnosis. This graphical analysis helps in verifying the initial success of the chosen model.

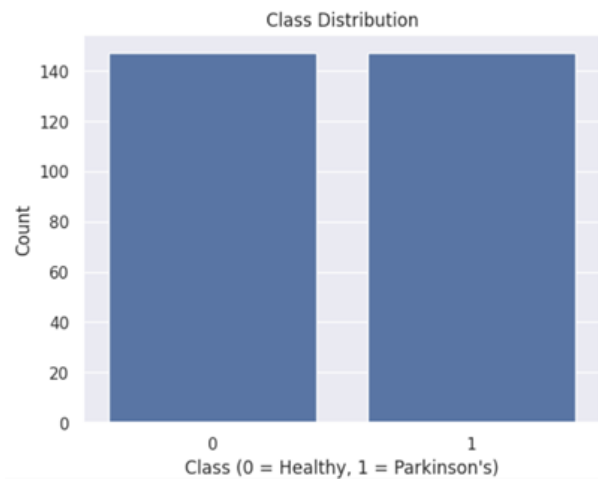


Fig 6. Class Distribution Bar Graph of Healthy Vs Parkinson's Infected

Figure 6 showcases a Class Distribution Bar Graph. The class distribution bar chart displays the proportion between the healthy class (Class 0) and Parkinson's class (Class 1) of the data set. Balanced data sets are necessary to avoid bias of the model towards the dominant class, causing over-prediction of common classes. Where the distribution is imbalanced, as may be suggested here, additional measures such as resampling or cost-sensitive learning would be required to ensure the model is fairer and more accurate.

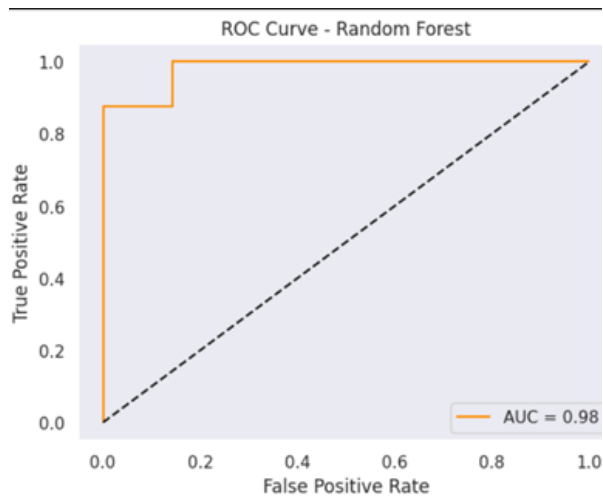


Fig 7. ROC Curve of Random Forest

Figure 7 illustrates the ROC Curve of the Random Forest Classifier. This ROC curve is unique to the Random Forest classifier. This much higher AUC value of 0.98 indicates that this model is performing very close to optimally compared to the others and can be a promising choice for detecting Parkinson's disease. Comparing this with AUC scores for other models, Random Forest looks to be the best, influencing the algorithm's choice for this task.

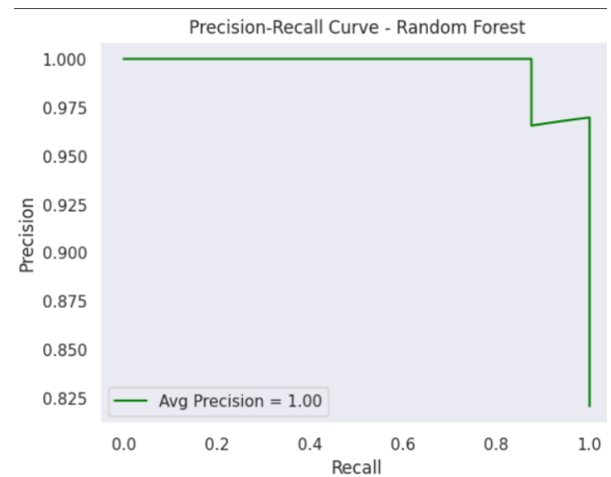


Fig 9. Precision-Recall Curve – Random Forest

Figure 9 shows a graph that illustrates the Precision-Recall Curve of the Random Forest Classifier. The precision-recall curve highlights the performance of the model in retrieving relevant cases, i.e., the correct identification of individuals affected by Parkinson's disease. The optimal average precision value of 1.00 indicates that the Random Forest classifier performs well in handling imbalanced datasets since it maximizes precision, which limits false positives, and recall, which limits false negatives. The measure has particular significance in handling medical datasets, where false negatives could provide severe consequences.

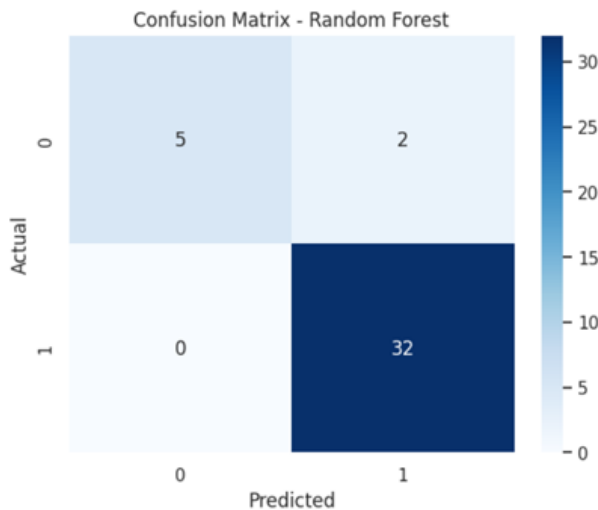


Fig 8. Confusion Matrix of Random Forest

Figure 8 showcases a Confusion Matrix based on the Random Forest Classifier. By identifying true positives, false positives, true negatives, and false negatives, the confusion matrix offers a comprehensive view of the model's performance. For the Parkinson's detection model, it is critical to limit false negatives since they represent undiagnosed cases. The matrix allows one to determine whether the Random Forest classifier correctly identifies cases of Parkinson's disease while excluding misclassification cases.

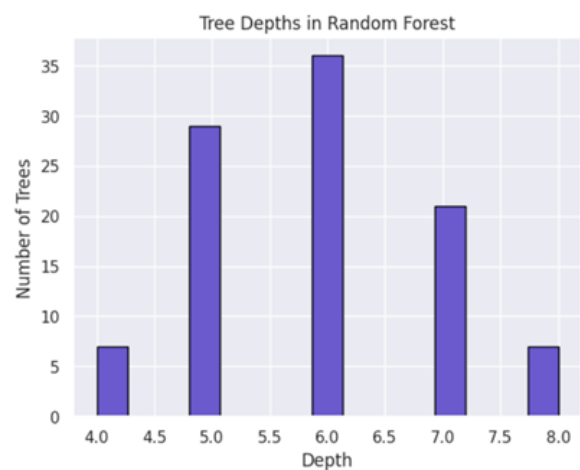


Fig10. Tree Depths in Random Forest

Figure 10 illustrates the Tree Depths in the Random Forest Classifier by using a bar graph. This graph provides an interpretation of the structural complexity of the Random Forest model. It shows the depth distribution of individual decision trees in the ensemble, which affects the model's ability to identify patterns in the data. Shallower trees may lead to oversimplification of the problem (underfitting), while very deep trees may lead to overfitting. Tracking the depth distribution is critical in achieving a balance between accuracy and generalizability.

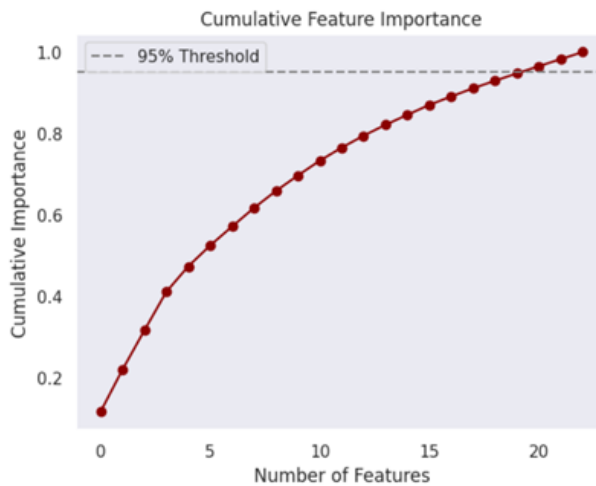


Fig 11. Cumulative Feature Importance

Figure 11 showcases a Cumulative Feature Importance using a line graph. This plot shows the effect of different features on the overall predictive ability of the model. Identification of the leading features that contribute to 95% of the performance allows the model to focus on the most informative data points, thus improving efficiency and interpretability. For a Parkinson's detection model, information on the significant features—e.g., variations in vocal pitch, tremor-related measurements, or clinical features—could provide significant information on the underlying biomarkers of the disease.

VI. CONCLUSION

The integration of Random Forest and K-Nearest Neighbours classifiers into the developed system provides an extensive platform for the precise identification of Parkinson's Disease through vocal biomarkers. Random Forest algorithm, given its nature of ensemble and decision trees based on entropy, provides enhanced robustness and interpretability and is thus suitable for handling high-dimensional and complex data. The KNN algorithm provides simplicity along with increased sensitivity towards local structures and thus provides for easy detection of subtle vocal changes in early Parkinson's Disease. These models, together, provide a robust platform for predictive diagnosis in neurodegenerative disease and towards the development of scalable, non-invasive screening tools for use in real-world clinical practice.

REFERENCES

- [1] Zhang, J., Zhang, W., Zhang, H., et al. (2023). Prediction of Parkinson's disease using machine learning methods. *Biomolecules*, 13(12), 1761.
- [2] Ji, Y., Zou, J., Zhang, J., et al. (2022). Machine learning models for diagnosis of Parkinson's disease using multiple structural magnetic resonance imaging features. *Frontiers in Aging Neuroscience*, 14, 633752.
- [3] Mei, J., Desrosiers, C., & Frasnelli, J. (2021). Machine learning for the diagnosis of Parkinson's disease: A review of literature. *Frontiers in Aging Neuroscience*, 13, 633752. *Frontiers+1arXiv+1*
- [4] Tabashum, T., et al. (2024). Machine learning models for Parkinson disease: Systematic review. *JMIR Medical Informatics*, 12, e50117. *JMI - JMIR Medical Informatics*
- [5] Mei, J., Desrosiers, C., & Frasnelli, J. (2020). Machine learning for the diagnosis of Parkinson's disease: A systematic review. *arXiv preprint, arXiv:2010.06101*.
- [6] Gunduz, H. (2019). Deep learning-based Parkinson's disease classification using vocal feature sets. *IEEE Access*, 7, 115540–115551.
- [7] Haq, A. U., Li, J. P., Memon, M. H., et al. (2019). Feature selection based on L1-norm support vector machine and effective recognition system for Parkinson's disease using voice recordings. *IEEE Access*, 7, 37718–37734.
- [8] Li, S., Lei, H., Zhou, F., et al. (2019). Longitudinal and multi-modal data learning for Parkinson's disease diagnosis via stacked sparse auto-encoder. In *2019 IEEE 16th International Symposium on Biomedical Imaging (ISBI 2019)* (pp. 384–387).
- [9] Liu, H., Du, G., Zhang, L., et al. (2016). Folded concave penalized learning in identifying multimodal MRI marker for Parkinson's disease. *Journal of Neuroscience Methods*, 268, 1–6.
- [10] Pahuja, G., & Nagabhushan, T. N. (2016). A novel GA-ELM approach for Parkinson's disease detection using brain structural T1-weighted MRI data. In *2016 Second International Conference on Cognitive Computing and Information Processing (CCIP)* (pp. 1–6).
- [11] Papadopoulos, A., Kyritsis, K., Klingelhoefer, L., et al. (2019). Detecting Parkinsonian tremor from IMU data collected in-the-wild using deep multiple-instance learning. *IEEE Journal of Biomedical and Health Informatics*, 24(9), 2559–2569.
- [12] Haller, S., Badoud, S., Nguyen, D., et al. (2013). Differentiation between Parkinson disease and other forms of Parkinsonism using support vector machine analysis of susceptibility-weighted imaging (SWI): Initial results. *European Radiology*, 23(1), 12–19.
- [13] Islam, M. S., Parvez, I., Hai, D., & Goswami, P. (2014). Performance comparison of heterogeneous classifiers for detection of Parkinson's disease using voice disorder (dysphonia). In *2014 International Conference on Informatics, Electronics & Vision (ICIEV)* (pp. 1–7).
- [14] Illán, I. A., Górriz, J. M., Ramírez, J., et al. (2012). Automatic assistance to Parkinson's disease diagnosis in DaTSCAN SPECT imaging. *Medical Physics*, 39(10), 5971–5980.
- [15] Peker, M. (2016). A decision support system to improve medical diagnosis using a combination of k-medoids clustering based attribute weighting and SVM. *Journal of Medical Systems*, 40(4), 11